

MADHURA PRATAP HANDE DESHMUKH

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SUMMARY

Embedded Systems Engineer with experience developing production-grade firmware for real-time, safety-critical platforms. Expertise in bare-metal firmware, low-level drivers, digital signal processing, and hardware-software integration.

EXPERIENCE

Noccarc Robotics Private Limited Firmware Engineer

Pune, India
June 2024 – present

- Built **production-grade firmware and embedded software** for **Patient Monitor NX5**, **Humidifier RH07** and **Ventilator V730i**, on STM32 ARM Cortex-M microcontrollers, contributing to commercial medical devices.
- Implemented **low-level drivers** and **hardware abstraction layers** for UART, SPI, ADC, DMA, GPIO, Timers, and external interrupts, enabling reliable interfacing with analog front ends (AFEs), sensors, displays, and peripheral devices.
- Engineered **DMA-driven** multi-channel data acquisition and **DSP pipelines** for physiological signal processing, achieving **98% measurement accuracy** while reducing processing **overhead** by **20%** across embedded monitoring applications.
- Resolved power management, battery charging, and hardware-software integration issues across embedded platforms, improving system reliability, **reducing downtime by 30%** while enabling reliable production deployment.

Embedded Software Intern

October 2023 – May 2024

- Developed MPU-based application software using queues, semaphores, mutexes, and timers for concurrent execution.
- Designed automated **PCB validation** and production test jigs, reducing manual testing effort by **84%**.

ACADEMIC PROJECTS

Design and Development of Balancing Robot

October 2022 - May 2023

- Engineered a self-balancing spherical robot achieving $\pm 2^\circ$ tilt accuracy, recovering from disturbances within **2 seconds**.

Robotic Hand Prosthesis

August 2021 - September 2022

- Developed an **EMG-driven** prosthetic hand, achieving lifelike finger articulation with response latency below **150 ms**.

Arrow Throwing Robot and Arrow Picking Robot

September 2020 - July 2021

- Developed an **autonomous swerve-drive** Arrow Throwing robot for ABU Robocon.

Rugby Ball Pick - Place Robot and Rugby Ball Kick Robot

August 2019 - June 2020

- Performed ball-kicking trajectory analysis in MATLAB, developed autonomous ball placing mechanism for ABU Robocon.

LEADERSHIP & ACTIVITIES

Project Lead | Ball Balancing Robot & Robotic Hand Prosthesis

Pune, India

- Ball Balancing Robot:** Led an 8-member multidisciplinary team through system design, development, integration, testing.
- Robotic Hand Prosthesis:** Directed the design, development and validation from concept to prototype.

Training & Placement Coordinator | Government College of Engineering and Research

Pune, India

TECHNICAL SKILLS

Programming: C, C++, Embedded C, Python **RTOS & Operating Systems:** FreeRTOS, CMSIS-RTOS, Linux, Windows
Embedded Software: Bare-metal, Firmware Architecture, Hardware-Software Co-Design, Digital Signal Processing
Microcontrollers: STM32 (ARM Cortex-M0/ M3/ M4), MSP430, ATmega
Hardware & Electronics: PCB Design, Board Bring-Up, Hardware Validation, Analog Front-End (AFE) Interfacing
Communication & Interfaces: UART, SPI, I2C, CAN, RS485, One-wire, ADC, DAC, DMA, PWM, GPIO, Timers, Interrupts
Development & Debugging: Keil uVision, Eclipse, VS Code, Git, Eagle, Oscilloscope, Logic Analyzer, JTAG, ST-Link
Languages: English (professional), Hindi (fluent), Marathi (native), Japanese (JLPT N4)

EDUCATION

Government College of Engineering and Research, Savitribai Phule Pune University

Pune, India

Bachelor of Engineering in Instrumentation and Control | GPA: 8.74/10

August 2019 - June 2023

AWARDS & HONORS

- Savitribai Phule Pune University Rank 10**, Instrumentation & Control Branch (2023)
- 2nd Place, COEP Tech I2I Innovation Challenge** (2022) for Robotic Hand Prosthesis
- ABU Robocon: **All India Rank 6** (2021), **All India Rank 14** (2020)